

Fixed or Dynamic Proximity-Based? Exploring the Impact of Syringe-Sharing Partnership Assumptions on Hepatitis C Virus Transmission Among People Who Inject Drugs Using Agent Based Modelling: An ODD Protocol

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1. Model Description

The model is an agent-based model (ABM) of hepatitis C virus (HCV) transmission among people who inject drugs (PWID). It is implemented in NetLogo and represents HCV transmission through syringe-sharing interactions within a population of PWID [1, 2]. The model was adapted from the network-based HIV transmission model developed by [3], with disease-specific parameters and epidemiological processes modified to reflect HCV transmission and treatment among PWID.

The model comprises interacting population, behavioural, transmission, treatment, and partnership components. Individual agents represent PWID and are assigned epidemiological states, treatment status, behavioural characteristics, and group membership. Transmission occurs through syringe-sharing interactions between agents, allowing infection to spread through the simulated population over time.

The primary purpose of the model is to compare two alternative representations of syringe-sharing partnerships: fixed partnerships and dynamic proximity-based partnerships. In the fixed-partnership formulation, agents maintain the same syringe-sharing partners throughout the simulation. In the dynamic proximity-based formulation, interaction opportunities depend on the spatial proximity of agents and may change over time. Apart from the partnership representation, all population characteristics, epidemiological processes, treatment assumptions, and transmission parameters remain identical across both formulations. The model therefore provides a controlled experimental framework for examining whether alternative assumptions regarding syringe-sharing partnerships influence simulated HCV transmission dynamics and the ability of the model to reproduce observed epidemiological patterns among PWID.

2. Purpose

The purpose of the model is to examine whether alternative assumptions regarding syringe-sharing partnerships influence simulated HCV transmission dynamics and the ability of the model to reproduce observed prevalence patterns among PWID.

3. Entities, State Variables and Scales

3.1 Entities

The model contains a single agent type representing people who inject drugs (PWID). Agents are characterised by epidemiological, behavioural, treatment, and partnership-related attributes that influence their risk of acquiring and transmitting HCV. In addition to PWID agents, the model includes entities that support interaction formation, spatial movement, and model execution. These entities and their roles within the model are summarised in Table 1.

Table 1. Entities represented in the partnership-assumption model.

Entity	Description
PWID agents	Individual agents representing people who inject drugs. Agents hold epidemiological, behavioural, treatment, and partnership-related attributes.
Links	Persistent syringe-sharing ties used only in the fixed partnership configuration. Links define the partners available for syringe-sharing exposure.
Patches	NetLogo spatial units used to position agents. In the dynamic proximity-based configuration, patches support local neighbourhood interactions.
Observer	The NetLogo observer manages global variables, simulation time, setup, annual recording, and export of transmission data.

3.2 State Variables

Each PWID agent is characterised by a set of epidemiological, behavioural, and network-related attributes. Epidemiological attributes describe the agent’s infection and treatment status. Behavioural attributes determine injecting and syringe-sharing activity, while network-related attributes define group membership and partnership relationships. Table 2 summarises the principal state variables used in the model.

Table 2. State variables assigned to PWID agents.

State variable	Description
susceptible?	Indicates whether the agent is susceptible to HCV infection.
chronic?	Indicates whether the agent has chronic HCV infection.
on-going-treatment?	Indicates whether the agent is currently receiving treatment.
cured?	Indicates whether the agent has cleared infection following treatment.
resistant?	Indicates treatment failure or resistant infection status.
syringe-sharer?	Indicates whether the agent belongs to the syringe-sharing population.
label	Group membership of the agent: core, inner, or outer.
injections-per-month	Monthly injecting frequency assigned to the agent.
injection-per-day	Daily number of injections sampled from the agent’s monthly injecting frequency.
CC_{it}	Number of critical contacts at the current time step.
R_{it}	Current risk factor derived from syringe-sharing exposure to infected partners.
tmp-total-partner-inj	Temporary count of total injections among partners encountered during the current day.
tmp-infectious-partner-inj	Temporary count of injections among infected partners encountered during the current day.
infectious-sampled-partners	List of infected partners sampled during the current day.
daily-proximity-partners	List of partners encountered through proximity during the current day.
was-infected-this-year?	Indicator recording whether the agent became infected during the current simulation year.
ever-started-treatment?	Indicator recording whether the agent has ever initiated treatment.
treatment-days	Number of days spent on the current treatment course.
treatment-rate	Annual probability of treatment initiation.
clearing-rate	Probability parameter used for natural clearance.
treatment-peer-effect	Peer-effect multiplier applied to treatment initiation probability.
contact-radius	Radius used to identify nearby partners in the dynamic proximity-based configuration.

3.2.1 Scales

The model operates using discrete daily time steps. During each time step, agents may interact with partners, experience transmission events, initiate treatment, or undergo treatment outcomes. Simulation outputs are aggregated annually to facilitate comparison with surveillance data and reported notification estimates.

The simulation is run for six years, corresponding to the period over which HCV prevalence trends are evaluated. The model is implemented within a two-dimensional NetLogo environment in which agents occupy spatial locations and move throughout the simulation space. The summary of the spatial and temporal parameters of the model is in Table3

Table 3. Spatial and temporal scales used in the model.

Scale	Description	Value
Spatial world	NetLogo two-dimensional patch world used to position and display agents.	NetLogo world
Patch coordinates	Coordinate range used in the interface.	−15 to 15
Basic time step	One model tick represents one simulated day.	1 day
Week	Aggregation of daily time steps.	7 days
Year	Aggregation of weekly time steps.	52 weeks = 364 days
Simulation horizon	Simulation continues until the year counter reaches 7.	6 completed years

4. Process Overview and Scheduling

The model operates using discrete daily time steps. The processes executed within the model is summarised in Table 4. During each time step, agents move within the simulation environment, establish syringe-sharing interactions according to the selected partnership assumption, and experience transmission and treatment processes. The sequence of events is identical across all simulation scenarios except for the mechanism used to identify interaction partners. This allows the effects of alternative partnership assumptions to be examined while keeping all other epidemiological and behavioural processes unchanged.

4.1 Daily Updates

At the beginning of each day, agents update their daily injection activity and reset temporary variables used to track interaction and exposure information. Agents then move within the simulation environment. Depending on the partnership configuration, syringe-sharing interactions are determined either through pre-existing network links (fixed partnerships) or through encounters with nearby agents within the local neighbourhood (dynamic proximity-based partnerships). Following interaction formation, treatment initiation is evaluated for eligible agents. Susceptible agents then calculate their exposure risk and may become infected through syringe-sharing interactions with infected partners. Finally, treatment outcomes are updated for agents currently receiving treatment.

4.2 Annual Updates

Model outputs are recorded annually. At the end of each simulation year, the model records the number of chronically infected agents, new infections, incidence, prevalence, treatment initiations, agents currently receiving treatment, and cured individuals. Following annual recording, infection flags used to identify new infections are reset in preparation for the next simulation year.

4.3 Stopping Condition

The model is initialised at year one and progresses through daily time steps until the end of year six. When the simulation reaches year seven, transmission data are exported and the simulation terminates.

5. Design Concepts

5.1 Basic Principles

The model is based on the network-based agent-based modelling framework developed by [3], in which disease transmission occurs through interactions between individual agents rather than through homogeneous population mixing. HCV transmission is represented as a consequence of syringe-sharing interactions between connected PWID. The model assumes that transmission risk is influenced by both the infection status of interaction partners and the frequency of injecting behaviour. The primary extension considered

Table 4. Main processes executed during each simulation time step.

Process	Description
Setup	Initialises the simulation environment, creates agents, assigns epidemiological states, behavioural attributes, and population groups, and establishes the partnership structure.
Daily injection update	Samples the number of injections performed by each agent during the current day and resets temporary exposure variables.
Movement	Agents move within the simulation environment according to the movement rules.
Partnership formation	Interaction partners are identified either through persistent network links (fixed partnerships) or through local neighbourhood encounters (dynamic proximity-based partnerships).
Risk calculation	Each susceptible syringe-sharing agent calculates its current exposure risk based on interactions with infected partners.
Treatment initiation	Eligible infected agents may initiate treatment according to the treatment process.
Transmission	Susceptible agents may acquire HCV infection through syringe-sharing interactions with infected partners.
Treatment outcome	Agents receiving treatment may complete treatment and transition to cured or resistant states.
Annual recording	Annual epidemiological and treatment outcomes are recorded.
Data export	Transmission events are exported when the simulation terminates.

in this model is the representation of syringe-sharing partnerships, which are implemented as either fixed or dynamic proximity-based interactions.

5.2 Emergence

Population-level outcomes emerge from the interactions of individual agents. Annual HCV prevalence, incidence, treatment uptake, and treatment outcomes are not imposed on the system but arise from the accumulation of syringe-sharing interactions, transmission events, treatment initiation, and treatment outcomes over time.

5.3 Adaptation

Agents do not adapt their behaviour in response to changes in infection prevalence, treatment availability, or epidemic conditions. Behavioural characteristics assigned to agents remain unchanged throughout the simulation.

5.4 Objectives

Agents do not possess explicit objectives or optimisation strategies. Their behaviour is governed by the epidemiological and behavioural rules specified within the model.

5.5 Sensing

Agents are able to identify potential interaction partners. In the fixed-partnership configuration, agents interact with individuals connected through established network links. In the dynamic proximity-based configuration, agents interact with nearby agents located within a specified neighbourhood radius.

5.6 Interaction

Interactions occur through syringe-sharing relationships. Depending on the experimental scenario, these relationships are represented either by persistent network links or by temporary proximity-based encounters. Infection can only occur through interactions involving infected partners.

5.7 Collectives

The PWID population is divided into three behavioural groups: core, inner, and outer. These groups represent different levels of syringe-sharing activity as well as interaction budgets per timestep and are used to capture heterogeneity in transmission risk within the population.

5.8 Stochasticity

Several processes are stochastic, including agent initialisation, movement, infection events, treatment initiation, treatment outcomes, natural clearance, loss of treatment-induced immunity, and partner selection in both partnership configurations. Consequently, repeated simulation runs may produce different epidemiological trajectories.

5.9 Observation

The model records annual epidemiological and treatment outcomes, including the number of chronically infected agents, new infections, incidence, prevalence, treatment initiations, cumulative treatment uptake, and cumulative cured individuals. In addition, transmission events are recorded to allow reconstruction of infection pathways and further network analysis.

6. Initialization

6.1 Population Structure

At model initialisation, a population of 1802 syringe-sharing PWID is created. This population size was derived from estimates of the syringe-sharing PWID population in Ireland and was used consistently across all simulation experiments. Each agent is assigned an initial epidemiological state, treatment status, and behavioural attributes. A proportion of agents are initialised as chronically infected to reflect the estimated HCV prevalence among Irish PWID at the beginning of the simulation period.

The composition of the the initial agent structure in the model is summarised in Table 5 To represent heterogeneity in syringe-sharing behaviour, agents are assigned to one of three behavioural groups: core, inner, or outer. These groups differ in their syringe-sharing activity and therefore contribute differently to transmission risk. The group composition is identical across all experimental scenarios, ensuring that differences in model outcomes can be attributed to the partnership assumptions being examined rather than differences in population structure.

Table 5. Initial population composition by behavioural group.

Group	Number of agents	Percentage of population
Core	94	5.2%
Inner	223	12.4%
Outer	1485	82.4%
Total	1802	100.0%

Agents are randomly positioned within the simulation environment at the start of each run. Infection status is then assigned by randomly selecting 985 agents to enter the simulation with chronic HCV infection, while all remaining agents are initialised as susceptible.

6.2 Global Parameters

The model uses a set of global parameters to define population size, infection prevalence, treatment processes, interaction levels, and partnership structure. These parameters remain fixed throughout each simulation run and were selected based on published literature, surveillance data, or assumptions inherited from the adapted modelling framework. Where appropriate, parameter values were modified to reflect HCV epidemiology among PWID in Ireland.

To ensure a fair comparison between partnership assumptions, the same parameter values were used in both the fixed-partnership and dynamic proximity-based scenarios. Consequently, any observed differences in model outcomes arise from the partnership representation rather than from differences in population, transmission, or treatment parameters. The global parameters used in the model are summarised in Table 6.

Table 6. Global variables used in the model.

Global variable	Description
day, week, year	Simulation clock variables used to track time.
shared-injections	Global variable used to store calculated shared injections.
annual-total-chronic	List storing annual counts of chronically infected agents.
annual-new-infections	List storing annual counts of new infections.
annual-incidence-rate	List storing annual incidence rates.
annual-prevalence-rate	List storing annual HCV prevalence rates.
annual-total-cured	List storing annual counts of cured agents.
annual-on-treatment	List storing annual counts of agents on treatment.
annual-ever-treated	List storing cumulative treatment initiation counts.
transmission-data	List of recorded infector–infected transmission pairs.
transmission-steps	List storing the simulation tick at which each transmission event occurred.
current-run-id	Identifier for the current simulation run.
run-counter	Counter used to assign run identifiers.
prob-syringe-sharing-core	Syringe-sharing probability assigned to core agents.
prob-syringe-sharing-inner	Syringe-sharing probability assigned to inner agents.
prob-syringe-sharing-outer	Syringe-sharing probability assigned to outer agents.
infection-rate	Transmission parameter used in the infection probability calculation.

6.3 Initial Conditions

At the beginning of each simulation, agents are assigned epidemiological states according to the initial prevalence assumptions. Individuals selected as infected enter the simulation with chronic HCV infection, while all remaining agents are initialised as susceptible. No agents begin the simulation receiving treatment, cured, or resistant.

The initial distribution of epidemiological states is identical across all simulation runs and partnership configurations. This ensures that comparisons between experimental scenarios are not influenced by differences in starting conditions. The initial epidemiological composition of the population is shown in Table 7.

7. Input Data

Model parameterisation was informed using a combination of Irish surveillance data and published literature describing HCV epidemiology, injecting behaviour, and treatment outcomes among PWID.

Table 7. Initial epidemiological state distribution at model initialisation.

Epidemiological State	Number of Agents	Percentage of Population
Susceptible	817	45.3%
Chronically infected	985	54.7%
On-going-treatment	0	0%
cured	0	0%
resistant	0	0%
Total	1802	100.0%

Input data were used to initialise the population, define epidemiological parameters, and calibrate the model to observed HCV prevalence patterns.

Population estimates were derived from Irish studies of PWID and surveillance reports. The total PWID population was estimated from national prevalence studies, while the syringe-sharing population used in the model was obtained by combining these estimates with evidence regarding the proportion of PWID who report syringe-sharing behaviour [4, 5]. Initial HCV infections were informed by Irish surveillance data and published estimates of HCV burden among PWID [4, 6]. These data were used to determine the initial epidemiological composition of the simulated population.

Disease-specific parameters were obtained from published studies of HCV natural history and treatment outcomes. Estimates of spontaneous viral clearance were informed by the work of [7], while treatment effectiveness was based on evidence describing the outcomes of direct-acting antiviral therapies [8]. Additional evidence regarding the contribution of injection drug use to HCV transmission was used to support assumptions regarding transmission within the syringe-sharing population [9].

Behavioural and interaction-related assumptions were adapted from the original modelling framework of [3]. In particular, the allocation of agents to core, inner, and outer behavioural groups, together with the distribution of interaction opportunities across these groups, followed the stylised network structure proposed in the original model. These assumptions were retained to provide a consistent basis for comparing alternative partnership representations.

The model interface parameters that were inputted at initialisation and their values are summarised in Table 8.

8. Submodels

This section describes the principal procedures implemented in the model. The submodels correspond to population initialisation, behavioural assignment, partnership formation, interaction generation, risk calculation, transmission, treatment, annual recording, and data export processes executed during a simulation run.

8.1 Setup and Global Initialisation

The setup procedure initialises the simulation environment, creates the PWID population, assigns epidemiological states, and establishes the selected partnership structure. Global counters and storage variables used to record annual prevalence, incidence, treatment outcomes, and transmission events are initialised. Simulation time is reset and the model clock is set to day one, week one, and year one.

8.2 Agent Creation and Infection Seeding

PWID agents are created and randomly positioned within the simulation environment. Each agent is assigned epidemiological attributes, treatment attributes, behavioural variables, and partnership-related variables. Agents are initially assigned to the susceptible state before a predefined number of individuals are randomly selected and assigned chronic HCV infection. These seeded infections represent prevalent infections present at model initialisation rather than infections generated during simulation.

Table 8. Interface parameters and calibrated model inputs.

Parameter	Value / option	Description
sharers-only-size	1802	Estimated syringe-sharing PWID population used in the main experiments.
initial-infected	985	Number of agents seeded with chronic HCV infection at model initialisation.
num-core	94	Number of syringe-sharing agents assigned to the core group.
num-inner	223	Number of syringe-sharing agents assigned to the inner group.
num-outer	1485	Number of syringe-sharing agents assigned to the outer group.
infection-rate	0.76	Transmission parameter used in the daily infection probability calculation.
treatment-rate	0.13	Annual treatment initiation probability assigned to agents.
clearing-rate	0.30	Natural clearance parameter used in the infection process.
treatment-peer-effect	0.20	Peer-effect multiplier used in treatment initiation.
contact-radius	1	Radius used to identify nearby partners in the dynamic proximity-based model.
debug-mode?	on/off	Controls whether diagnostic information is printed during simulation.

8.3 Syringe-Sharing and Group Assignment

A predefined number of agents are designated as syringe sharers. Syringe-sharing agents are subsequently allocated to one of three behavioural groups: core, inner, or outer. Group assignment determines the syringe-sharing behaviour of each individual and provides the behavioural heterogeneity represented in the model. Any remaining syringe-sharing agents not explicitly assigned to the core or inner groups are allocated to the outer group.

8.4 Partnership Formation

Partnership formation depends on the experimental scenario being examined.

In the fixed-partnership configuration, syringe-sharing partnerships are represented by persistent links between agents. Links are created during model initialisation and remain unchanged throughout the simulation. Separate link-generation procedures are used to allocate the specified numbers of core, inner, and outer partnerships.

In the dynamic proximity-based configuration, no permanent syringe-sharing links are created. Instead, interaction opportunities are generated during each simulation day based on spatial proximity between agents.

8.5 Movement and Interaction Generation

At each daily time step, agents move within the simulation environment. Under the dynamic proximity-based configuration, movement alters the local neighbourhood of each agent and therefore influences which individuals are available for interaction.

Interaction opportunities are generated according to predefined interaction budgets allocated to the core, inner, and outer groups. During each interaction event, a potential partner is selected from agents

located within the specified contact radius. Information regarding interaction partners and their injecting behaviour is recorded for subsequent risk calculations.

8.6 Risk Calculation and Transmission

The risk factor (R_{it}) and probability of infection $P(\text{infection}_{it})$ are calculated following [3]. For each susceptible syringe-sharing agent, the model calculates a daily R_{it} based on the agent's syringe-sharing probability and the infection status of its interaction partners.

Under the dynamic proximity-based configuration, exposure is calculated using the partners encountered during the current daily proximity interaction step. The risk factor is given by:

$$[R_{it} = p_i \times \frac{I_{it}^P}{T_{it}^P}]$$

where (p_i) is the syringe-sharing probability assigned to agent (i) based on it's group, (I_{it}^P) is the number of daily injections associated with infected proximity partners, and (T_{it}^P) is the number of daily injections associated with all proximity partners encountered by agent (i) during time step (t). If no proximity partners are encountered, ($R_{it} = 0$).

Under the fixed-partnership configuration, exposure is calculated using the agent's persistent link-neighbours. The model first calculates the total number of daily injections among linked partners and the weighted number of injections contributed by infected linked partners. The risk factor is then calculated using the same conceptual form:

$$[R_{it} = p_i \times \frac{I_{it}^L}{T_{it}^L}]$$

where (I_{it}^L) is the weighted number of injections among infected linked partners and (T_{it}^L) is the total number of injections among all linked partners. If an agent has no linked partners or if partner injections are zero, ($R_{it} = 0$).

The number of critical contacts for agent (i) at time (t) is then calculated as:

$$[CC_{it} = x_{it} * R_{it},]$$

where (x_{it}) is the number of injections performed by agent (i) on day (t).

The probability that a susceptible agent becomes infected during the current day is:

$$[P(\text{infection}_{it}) = 1 - \left(1 - \frac{\beta}{364}\right)^{CC_{it}}]$$

where (β) is the annual transmission parameter. Infection is determined by a Bernoulli trial using this probability. When infection occurs, the agent is assigned chronic HCV infection and the infector-infected pair is recorded where an infectious partner can be identified.

When infection occurs, the agent transitions from the susceptible state to the chronic infection state. The model records the source of infection by identifying an infected interaction partner and storing the transmission event for subsequent analysis. Newly infected agents may also experience spontaneous viral clearance according to the specified clearance probability.

8.7 Treatment

Chronically infected or resistant agents who are not already receiving treatment are eligible to initiate treatment. Because the treatment parameter is specified as an annual probability, it is converted into a daily probability:

$$[p_{\text{daily}} = 1 - (1 - p_{\text{annual}})^{1/364}]$$

where (p_{annual}) is the annual treatment initiation probability and 364 is the number of days in one model year.

Treatment initiation is also influenced by the treatment status of the agent's interaction partners. The final daily probability of starting treatment is calculated as:

$$[p_{\text{start}} = \min(1, p_{\text{daily}}(1 + \alpha q_{it}))]$$

where (α) is the treatment peer-effect parameter and (q_{it}) is the proportion of the agent's current partners who are receiving treatment.

Agents who initiate treatment remain on treatment for 84 days. At the end of this treatment period, treatment succeeds with probability 0.95. Successful treatment moves the agent to the cured state, while

treatment failure moves the agent to the resistant state. Cured agents may subsequently return to the susceptible state through loss of treatment-induced immunity, allowing reinfection to occur.

8.8 Annual Recording

At the end of each simulation year, epidemiological and treatment outcomes are recorded. These include the number of chronically infected individuals, cumulative treatment initiations, individuals currently receiving treatment, cured individuals, annual incidence, and annual prevalence. Annual infection indicators are then reset in preparation for the following year.

8.9 Transmission Recording and Data Export

Each transmission event is recorded together with the identities of the infector and infected agents and the simulation time step at which transmission occurred. Upon completion of the simulation, transmission records are exported to a comma-separated values (CSV) file to support subsequent network and transmission-pathway analyses.

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